

Subject Description Form

Subject Code	LSGI2802
Subject Title	Fundamentals of Urban Science
Credit Value	3
Level	2
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	The objectives of this subject are: • To provide theories and methodologies for urban science; the theories of urban systems, including their form, function, and management. • To provide concepts and theories for urban technologies, how they emerge from existing disciplines, and what they can do in smart urban management. • To equip students with knowledge of methodological skills to work with different urban models, and to apply these models to understand and address real-world challenges in the urban context.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Demonstrate an understanding of the basic theories about how urban systems are shaped and transformed; b. Explain the basic propositions of urban models, and what they tell us (and don't tell us) about complex urban systems; c. Reflect on the applications of urban technologies for contemporary cities and smart urban management; d. Understand the role of urban models in the urban science domain and their applications in real-world scenarios; e. Use different techniques to effectively visualize and communicate outcomes from urban models.
Subject Synopsis/ Indicative Syllabus	• Introduction to Urban Science: an overview of urban science including an introduction to complex dynamic urban system, the interactions between different urban sub-systems, as well as the contributions of urban economics, planning, and geomatics technologies to this area of study. • Urban Structure and Systems: the city as a geographical entity; the evolution of urban structure; and the linkages between

	different urban sub-systems (e.g., the linkages between energy, transport and the environment). • Exploring Urban Issues: application of urban science concepts to explore topics such as centrifugal (expansion and sprawl) and centripetal (agglomeration) urban growth; urban design and housing; citizen participation and governance; and urban policy, and management in the age of smart cities. • Urban Technologies and Models: introduction to different urban technologies and models, as well as their applications in urban management, including urban sensing technologies, network analysis, and urban simulations (e.g., spatial interaction, cellular automata, agent-based simulation), and urban economic and demographic forecasting. • Urban Science and Future Cities: reflecting on the role of urban science in smart urban planning and management for future cities, from a sustainable perspective.																																								
Teaching/Learning Methodology	Students will obtain the theories and methodologies in normal lectures. Students will then gain the practical experience through well-designed tutorial sessions. A group project will then follow to develop students' high-level cognitive understanding and integration of knowledge, and develop their teamwork and presentation skills.																																								
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. Assignment 1</td><td>30%</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Assignment 2</td><td>30%</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Group Project</td><td>40%</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100 %</td><td colspan="5"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Continuous assessment will be used to assess students' basic understanding of the theories, methods, and techniques in urban science. In assignment 1, students will be asked to deliver a short essay reviewing the theoretical developments of urban science and complex urban systems. Assignment 2 will ask students to choose one or more urban models and then to write an essay introducing their	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					a	b	c	d	e	1. Assignment 1	30%	√	√	√	√	√	2. Assignment 2	30%	√	√	√	√	√	3. Group Project	40%	√	√	√	√	√	Total	100 %					
Specific assessment methods/tasks	% weighting			Intended subject learning outcomes to be assessed (Please tick as appropriate)																																					
		a	b	c	d	e																																			
1. Assignment 1	30%	√	√	√	√	√																																			
2. Assignment 2	30%	√	√	√	√	√																																			
3. Group Project	40%	√	√	√	√	√																																			
Total	100 %																																								

	applications in different domains (e.g., urban transportation, urban governance, urban crime and security, urban energy, etc.). In addition, students will engage in a group project culminating in a presentation, where they will demonstrate their knowledge of urban science related to particular urban issues (e.g., human mobility). They will use advanced software tools to develop urban models and present their models and outcomes in class.	
Student Study Effort Expected	Class contact:	
	▪ Lecture	26 Hrs.
	▪ Tutorial	13 Hrs.
	Other student study effort:	
	▪ Self-study, reading and revision	39 Hrs.
	▪ Assignments and projects	39 Hrs.
	Total student study effort	117 Hrs.
Reading List and References	Shi, W.Z., Goodchild, M.F., Batty, M. Kwan, M.P., and Zhang, A.S. (2021). Urban Informatics, Springer. Batty, M. (2022). The Linear City: illustrating the logic of spatial equilibrium. Computational Urban Science, 2(1), 1-17. Portugali, J. (Ed.). (2021). Handbook on Cities and Complexity. Edward Elgar Publishing. Acuto, M., Parnell, S., & Seto, K. C. (2018). Building a global urban science. Nature Sustainability, 1(1), 2-4. Batty, M. (2018). Inventing future cities. MIT press. Foot, D. (2017). Operational urban models: an introduction. Routledge. Hall, T., & Barrett, H. (2012). Urban geography. Routledge. Balchin, P. N., Isaac, D., & Chen, J. (2019). Urban economics: a global perspective. Bloomsbury Publishing.	